Tables, Graphs & Constant Change

Four questions over the first two pages: solve, diamond, and two spirals.

TEACHER COPY | Display student pages; keep solutions private

1 | Solve

Solve for x. Show each step, then check your answer.

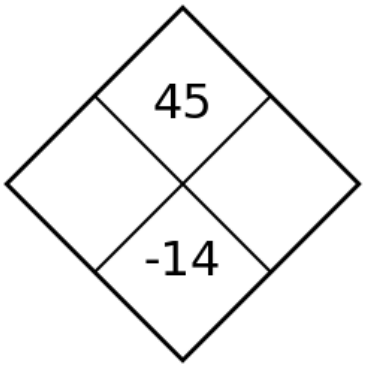
$$2(3x + 1) - 4 = 22$$

ANSWER / TEACHING NOTE

2(3x + 1) - 4 = 22 → 6x + 2 - 4 = 22 → 6x = 24 → x = 4.

2 | Diamond

Find two numbers whose product is at the TOP and sum is at the BOTTOM.



SOLUTION: -9 and -5 (product 45, sum -14).

Spiral Retrieval

Retrieve ideas from earlier Algebra 1 work. Explain your reasoning.

3 | Spiral Review

Is {(-1,4), (0,4), (2,4)} a function? Why?

ANSWER / TEACHING NOTE

Yes: each input has exactly one output; repeated outputs are allowed.

4 | Spiral Review

List the domain and range of {(2,7), (4,9), (6,7)}.

ANSWER / TEACHING NOTE

Domain {2,4,6}; range {7,9}.

What We Need to Know

Tables, Graphs & Constant Change

Learning focus | LG 02-ican-4 / LG 03-ican-1

- I can recognize constant additive change in a table.
- I can compare vertical change with horizontal change.
- I can connect a table, graph, and equation that describe the same linear pattern.

Vocabulary | Highlight the important words

Linear relationship

A relationship that has a constant rate of change and graphs as a straight line.

Constant difference

The same change in output for each equal change in input.

Horizontal change (Delta x)

The difference between x-coordinates of two points; movement left or right.

Vertical change (Delta y)

The difference between y-coordinates of two points; movement up or down.

Slope

The ratio of vertical change to horizontal change, Delta y / Delta x.

Starting value

The output at input zero; where a linear graph crosses the y-axis.

Meaning & Method | Rewritten review notes

1. Compare equal input intervals when checking for a linear pattern. A constant output difference is meaningful only when the input step is constant.

2. Compute slope as change in y divided by change in x. Keep the SAME point order in numerator and denominator.

3. For a linear rule y = mx + b, m is the slope and b is the output when x = 0.

These are original, concise review notes aligned to the course lessons; they are not a reproduction of the textbook pages.

Examples & You Try Its

Complete the Example with your teacher. Then solve the matching You Try It.

Example

Find the rate and equation represented by this table.

x	y
0	2
2	8
4	14

Record the method and steps:

> **ANSWER / TEACHING NOTE**
> Δx = +2, Δy = +6; m = 6/2 = 3; initial output b = 2.
> Equation y = 3x + 2.

You Try It

Follow the same steps to identify slope and an equation.

x	y
0	11
2	7
4	3

Find Delta x, Delta y, slope, and equation.

> **ANSWER / TEACHING NOTE**
> Δx = 2; Δy = -4; m = -4/2 = -2; initial output b = 11.
> Equation y = -2x + 11.

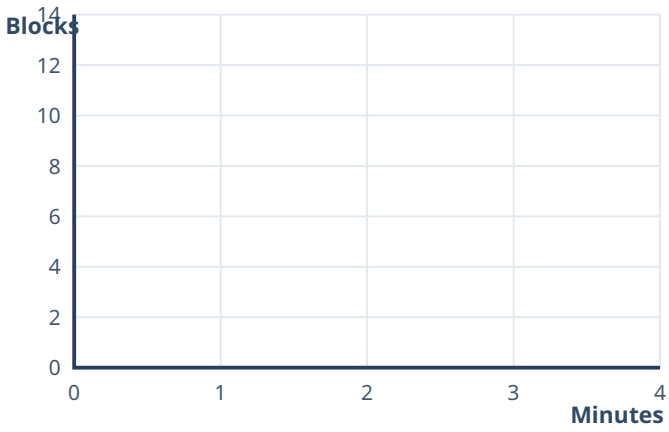
Explain or check your approach

Launch | One story, three views

Notice. Try it. Explain your reasoning with a partner.

A walker starts 2 blocks from school and moves 3 more blocks away each minute. Complete the table and sketch the points; connect them if time is continuous.

Time (min)	Blocks from school
0	2
1	___
2	___
3	___



1 | How is the +3 visible in the table and on the graph? Write a rule using distance d and time t.

ANSWER / TEACHING NOTE

Table outputs: 2, 5, 8, 11. Graph these points. Rise = 3 for run = 1. Equation d = 3t + 2.

2 | A different walker starts 5 blocks away and moves 2 blocks each minute. Compare their changes for one minute (do not solve where they meet).

ANSWER / TEACHING NOTE

Walker 1 rate 3 blocks/min versus Walker 2 rate 2 blocks/min.

Partner share: Compare reasoning, not just answers. What would convince someone else your conclusion is correct?

Review & Preview

Independent practice. A star marks a useful quick diagnostic for the November 10 test.

1 Find the slope of a table with x-values 0,2,4,6 and y-values 6,12,18,24. Explain each change.

ANSWER / TEACHING NOTE

Δy = +6 for Δx = +2, so slope = 3.

2 A line passes through (0,6), (2,10), and (4,14). Find its slope and starting value.

ANSWER / TEACHING NOTE

Slope = (10 - 6)/(2 - 0) = 2; starting value = 6.

3 ★ The table (0,8), (1,5), (2,2), (3,-1) is linear. Find slope and write the rule.

ANSWER / TEACHING NOTE

m = -3 and b = 8, so y = -3x + 8.

Review & Preview

Independent practice. A star marks a useful quick diagnostic for the November 10 test.

4 ★ Compare: Pattern A adds 2 to the output for each +1 input. Pattern B adds 4. Which graph is steeper? Explain.

ANSWER / TEACHING NOTE

B is steeper: its positive slope 4 is greater than A's positive slope 2.

5 Spiral: Is {(1,2),(1,2),(3,5)} a function? State the domain and range.

ANSWER / TEACHING NOTE

Yes; repeating exactly (1,2) does not violate the function rule. Domain {1,3}; range {2,5}.

6 Preview: A runner gains 18 meters in 3 seconds. What is the average rate of change, including units?

ANSWER / TEACHING NOTE

18 m / 3 s = 6 m/s.

Evidence check: Star a question you can explain confidently. Circle one you want to revisit on Monday, November 9.